Intrinsic invariants of $(3,4)$-cuspidal edges

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JA, surfaces and beyond

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   - Motivation
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3. new framework
   - ~~NACS~~
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new intrinsic invariants & Main results

the 1st invariant (β & comb. ty)

the 2nd invariant (σ & K.)

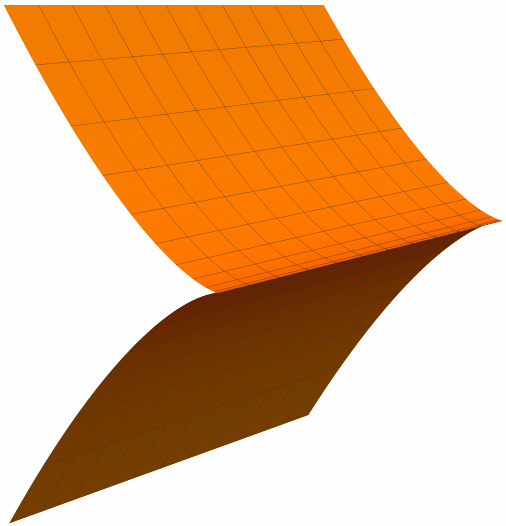
Cuspidal edge and $(3,4)$-cuspidal edge

Definition

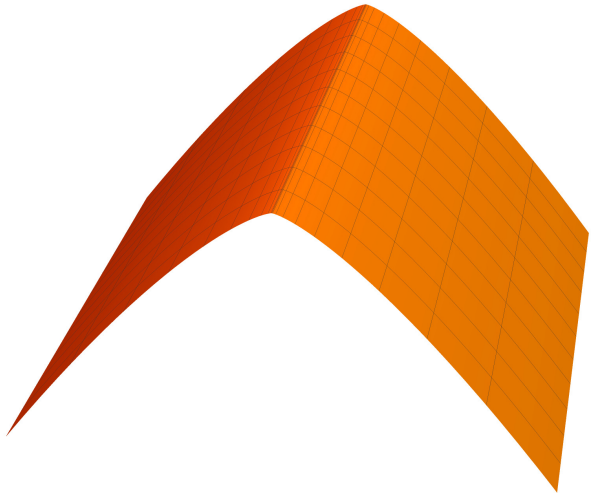
Let $f : \Sigma \to \mathbb{R}^3$ be a smooth map from a two-dimensional manifold Σ.

(i) f has a **cuspidal edge** at $p \in \Sigma$ if $f : (\Sigma, p) \to (\mathbb{R}^3, f(p))$ is $\mathcal{A}$-equivalent to $\boxed{(u, v) \longmapsto (u,\ v^2,\ v^3).}$

(ii) f has a **(3,4)-cuspidal edge** at $p \in \Sigma$ if $f : (\Sigma, p) \to (\mathbb{R}^3, f(p))$ is $\mathcal{A}$-equivalent to $\boxed{(u, v) \longmapsto (u,\ v^3,\ v^4).}$



Cuspidal edge



$(3,4)$-cuspidal edge

Cuspidal edges and $(3,4)$-cuspidal edges

- Both cuspidal edges and $(3,4)$-cuspidal edges arise naturally on **parallel surfaces**.

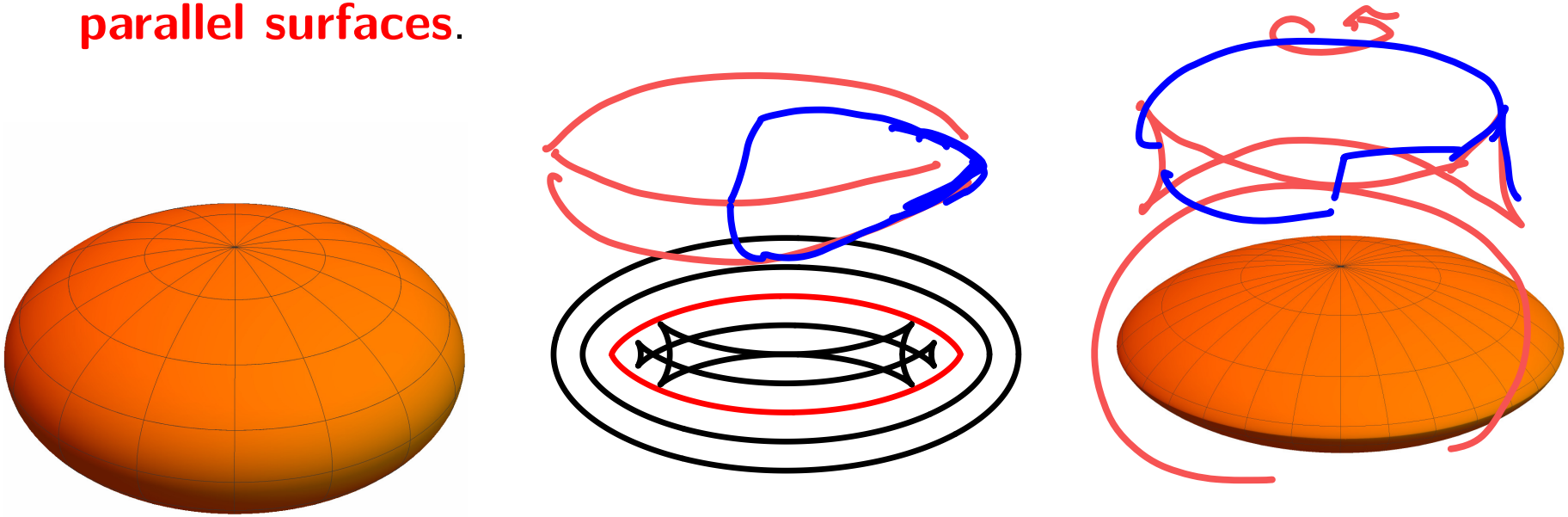


Figure: A rotational ellipsoid (left), its parallel curves (center), and a parallel surface (right). The red curve has a $(3,4)$-cusp.

- However, the geometry of $(3,4)$-cuspidal edges is less understood than that of ordinary cuspidal edges.

Previous studies on cuspidal edges

[KRSUY]

- **Criteria for cuspidal edges**
  Kokubu–Rossman–Saji–Umehara–Yamada (2005)

- **Singular curvature and geometry of cuspidal edges**
  Saji–Umehara–Yamada (2009)

[SUY]

- **Cuspidal curvature, product curvature, and Gaussian curvature**
  Martins–Saji–Umehara–Yamada (2016)

[MSUY]

- **Isometric deformations of cuspidal edges**
  Naokawa–Umehara–Yamada (2016),
  Honda–Naokawa–Umehara–Yamada (2020)

[NUY]
[HNUY]

boundedness of Gaussian and mean curvatures

[HNSUY] Honda-Naokawa-Saji-Ume-Yan (2020)

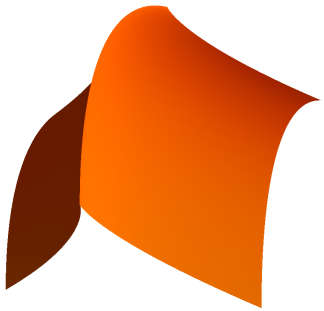
Introduction: Difference from cuspidal edges

Singular curvature (Saji–Umehara–Yamada (2009))

The singular curvature κ_s is an intrinsic invariant, and its sign is well-defined.

$$\operatorname{sgn}\kappa_s \quad \leftrightsquigarrow \quad \text{local shape}$$

$\kappa_s > 0$



$\kappa_s < 0$



$(3,4)$-cuspidal edge

The sign of κ_s depends on the orientation of the null vector field:

$$\eta \longmapsto -\eta \quad \Longrightarrow \quad \kappa_s \longmapsto -\kappa_s.$$

Hence its sign is not well-defined.

Research question

For $(3,4)$-cuspidal edges, the sign of κ_s is not well-defined

Question

Can we construct intrinsic invariants that describe the geometry of $(3,4)$-cuspidal edges?

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For $(3,4)$-cuspidal edges, the sign of κ_s is not well-defined

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Can we construct intrinsic invariants that describe the geometry of $(3,4)$-cuspidal edges?

We obtain some results:

Local shape

When $(3,4)$-CE is convex or saddle

Gaussian curvature

When K is bounded

Intrinsic criteria

Characterize Singularity type



Basic setting

Let $f : \Sigma \to \mathbb{R}^3$ be a smooth map from a two-dimensional manifold Σ.

Definition(Singular point)

p is a **singular point** if $\mathrm{rank}(df_p) < 2$.
The **singular set** is

$$S(f) := \{q \in \Sigma \mid \mathrm{rank}(df_q) < 2\}.$$

3-type edges: a class containing $(3,4)$-cuspidal edges

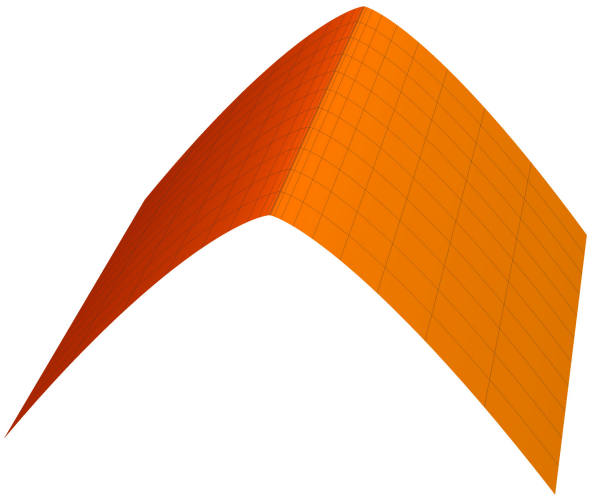
Definition(3-type edges) Martins–Saji–Santos–Teramoto (2024)

$f : \Sigma \to \mathbb{R}^3$ has a **3-type edge** at $p \in \Sigma$ if the map-germ $f : (\Sigma, p) \to (\mathbb{R}^3, f(p))$ is $\mathcal{A}$-equivalent to

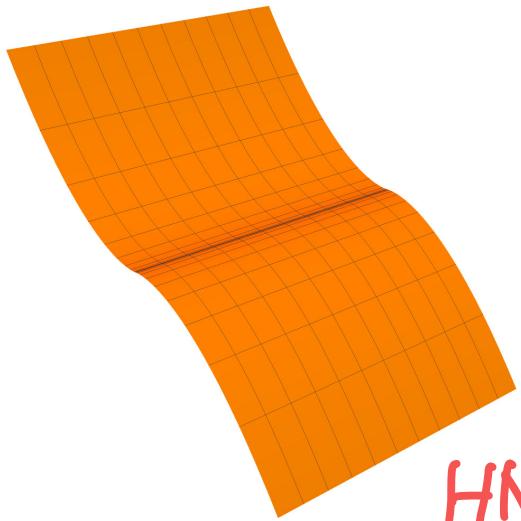
$$(u, v) \longmapsto \left(u,\ v^3,\ v^4\rho(u, v)\right)$$

for some smooth function ρ.

(u, v^3, v^4): $(3,4)$-cuspidal edge,



(u, v^3, v^5): $(3,5)$-cuspidal edge.



HNSUY

~~2-type edge~~ $(u, v) \longmapsto \left(u,\ v^2,\ v^3\rho\right)$ is called generalized cuspidal edge (2020))

Basic setting

Definition (Frontal)

f is a **frontal** if, for every $p \in \Sigma$, there exist a neighborhood U of p and a smooth map $\nu : U \to \mathbb{S}^2$ such that

$$\langle df_q(X), \nu(q) \rangle = 0 \quad (q \in U,\ X \in T_q\Sigma).$$

Basic properties of 3-type edges

If f has a 3-type edge at p, then

(i) $S(f)$ is a regular curve near p. ← rank 1

(ii) f is a frontal near p.

NACS

In this talk, we write $\boxed{ds^2 = E\,du^2 + 2F\,du\,dv + G\,dv^2}$, $\boxed{G_{v^j} := \dfrac{\partial^j G}{\partial v^j}}$.

Normalized adapted coordinate system (NACS)

A local coordinate system (u, v) is called a **NACS** if

- $S(f) = \{v = 0\}$,
- $f_v(u, 0) = f_{v^2}(u, 0) = \mathbf{0}$,
- $\{\, f_u(u, 0),\ f_{v^3}(u, 0),\ \nu(u, 0)\,\}$ is an orthonormal basis of $\mathbb{R}^3$ and

$$E(u, 0) = 1, \qquad F(u, v) \equiv 0,$$
$$G(u, 0) = G_v(u, 0) = G_{vv}(u, 0) = G_{v^3}(u, 0) = 0,$$
$$G_{v^4}(u, 0) = 6.$$

generalized ($\in$ (2-type edge))
2つNACSが存在
すること [HHNSUY 2015]

Existence

Every 3-type edge admits a NACS locally.

Remark : NACS for CE

full

Fact (HHN(U) 2017 (generalized) CE ≅ ZFC)

$\exists (u.v)$ s.t.

$E(u0) = 1$

$F \equiv 0$

$G(u.0) = G_v(u.0) = 0$

$G_{v^2}(u.0) = 2$

Intrinsic invariant β

We introduce our first new invariant.

Definition (β)

Let p be a 3-type edge of f, and choose a NACS (u, v) centered at p. Define

$$\beta(p) := \left[E_{v^4} + \frac{G_{v^5}}{20} E_{v^3} \right]_{(0,0)}.$$

For another NACS centered at p,

$$\widehat{\beta}(p) = \beta(p).$$

So, β is intrinsic invariant (determined by the first fundamental form).

Main result I: intrinsic detection of the local shape

Let f have a 3-type edge at p. The limiting tangent plane is

$$LT_p := f(p) + \nu(p)^{\perp}.$$

Defitnition (Convex and saddle-shaped)

Let LT_p be the limiting tangent plane at p.
f is **convex at** $p \iff \exists$ neighborhood V of p such that

$$f(V) \cap LT_p = \{f(p)\},$$

15 min

and $f(V) \setminus \{f(p)\}$ lies on one side of LT_p.

f is **saddle-shaped at** $p \iff$ for every neighborhood V of p, $f(V)$ meets both sides of LT_p.

Main result I: intrinsic detection of the local shape

Proposition A

Let $f : \Sigma \to \mathbb{R}^3$ be a frontal, and let $p \in \Sigma$ be a $(3,4)$-cuspidal edge. Then

- $\boxed{\beta(p) > 0 \implies f \text{ is convex at } p}$
- $\boxed{\beta(p) < 0 \implies f \text{ is saddle-shaped at } p}$

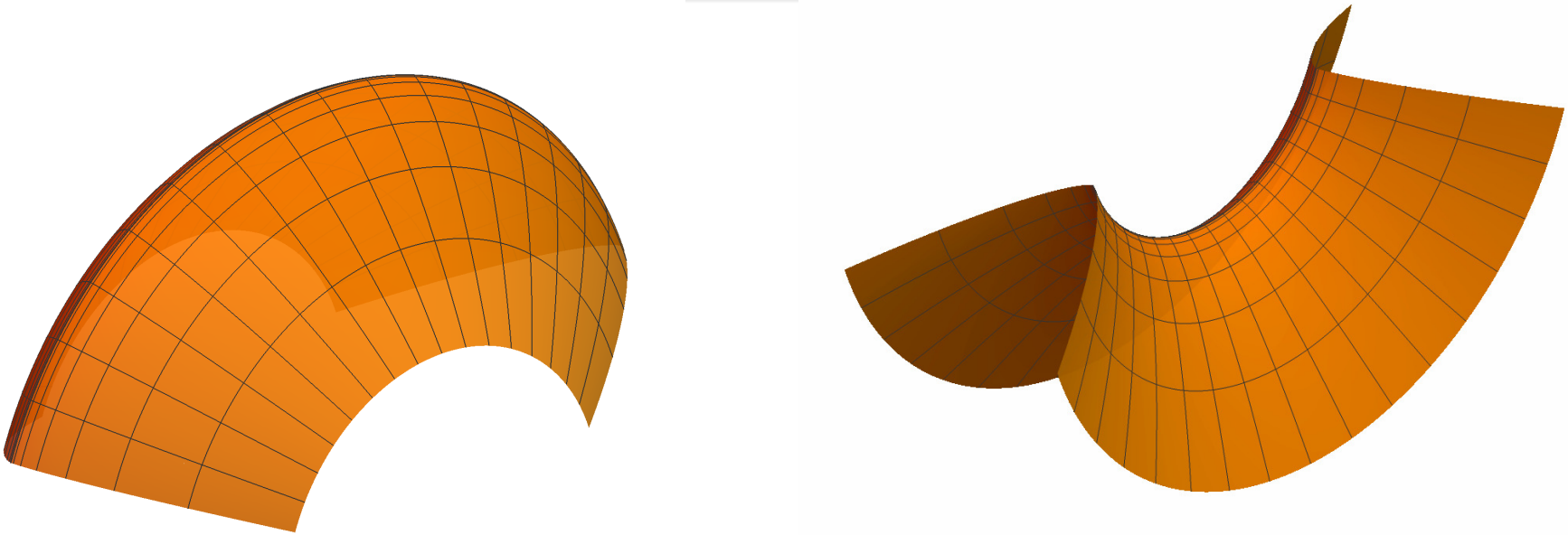


Figure: Local shapes at p: $\beta(p) > 0$ (left: convex) and $\beta(p) < 0$ (right: saddle-shaped).

17 min

Another intrinsic invariant δ

We next introduce a second intrinsic invariant to study Gaussian curvature.

Definition (δ)

Let p be a 3-type edge of f, and choose a NACS (u, v) centered at p. Define

$$\delta(p) := \left[-E_{v^5} + \frac{G_{v^5}}{12} E_{v^4} - \left(\frac{7G_{v^5}^2}{1200} - \frac{G_{v^6}}{30} \right) E_{v^3} \right]_{(0,0)}.$$

For another NACS centered at p,

$$\widehat{\delta}(p) = \pm\delta(p).$$

So, δ is also intrinsic invariants.

What do β and δ measure?

Let K be the Gaussian curvature on the regular set.

Expansion of $\nu^2 K$

In a NACS,

$$\nu^2 K = \frac{1}{3}\beta + \left(\frac{1}{6}\delta - \frac{G_{\nu^5}}{360}\beta\right)\nu + \mathcal{O}(\nu^2).$$

$\beta \neq 0$

At a fixed point of the singular set,

$$K = \frac{\beta(u)}{3\nu^2} + \mathcal{O}(\nu^{-1}).$$

β controls the leading divergence.

$\beta \equiv 0$

If $\delta \neq 0$, then

$$K = \frac{\delta(u)}{6\nu} + \mathcal{O}(1).$$

δ controls the next divergence.

Main result II: boundedness of Gaussian curvature

Theorem B

Let $f : \Sigma \to \mathbb{R}^3$ be a frontal, and let $p \in \Sigma$ be a 3-type edge.
Then, the following conditions are equivalent:

(i) There exists a neighborhood U of p such that K is bounded on $U \setminus S(f)$.

(ii) There exists a neighborhood U of p such that
$\boxed{\beta \equiv 0, \quad \delta \equiv 0 \quad \text{on } S(f) \cap U.}$

- We can detect blow-up of K intrinsically.
- Moreover, bounded K extends smoothly across $S(f)$.

次にスライドを追加して、カスプ辺の場合をコメント
$\nu K = K_\pi t + O(\nu)$

- MSOF で $K_T = K_V K_C$ (product curvature)
- generalized で
  $\forall K = K_{T1}(u) + O(V)$
- K: bdd $\Longleftrightarrow$ $K_{T1}(u) \equiv 0$ (MSOF)

3-type edge のとき.

$$\forall^2 K = \beta + (\quad)\forall + O(\forall^2)$$

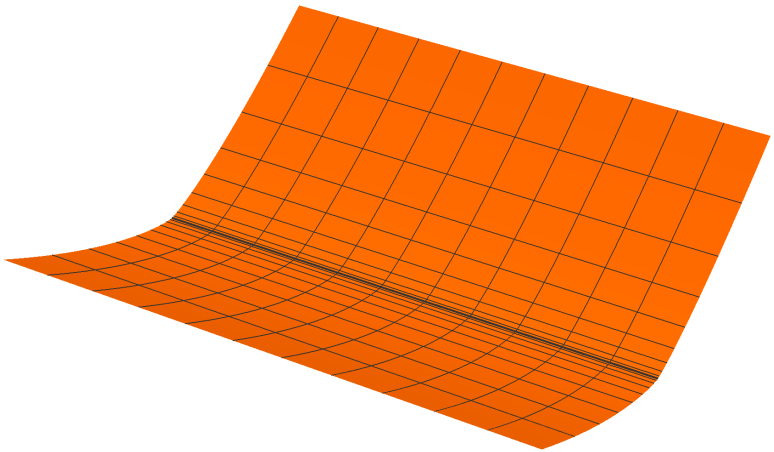
$\beta = K_T W_{23}$ (product で表示できる)

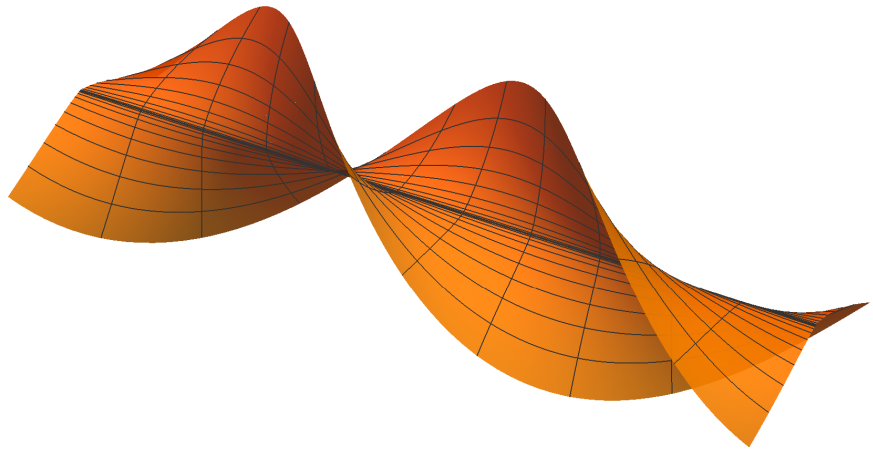
δ は product の表示を出てない.

($\beta \equiv 0$ かつ K_V 0のとき. 残存できる)

Examples of main result II

Both ((3, 4)-cuspidal edge) examples satisfy $\boxed{\beta \equiv 0, \quad \delta \equiv 0.}$





$$f_0(u,v) = \left(u, \frac{v^3}{6}, \frac{v^4}{24}\right)$$

$\boxed{K \equiv 0}$

$$f_{-1}(u,v) = \left(u - \frac{v^7}{1008}, \frac{v^3}{6}\cos u - \frac{v^4}{24}\sin u, \frac{v^3}{6}\sin u + \frac{v^4}{24}\cos u\right)$$

25min

$\boxed{K \longrightarrow -1 \qquad (v \to 0)}$

Main result III: intrinsic criteria for singularity types

Let $\gamma(0) = p$ parametrize $S(f)$, and set $\hat{\gamma} := f \circ \gamma$.
The osculating plane of $\hat{\gamma}$ at p is

$$OP_p := f(p) + \mathrm{span}\{\hat{\gamma}'(0), \hat{\gamma}''(0)\},$$

Definition (Generic) (cf. SUY 2009)

p is **generic** if the osculating plane of $\hat{\gamma}$ exists and differs from LT_p.

Equivalently, for 3-type edge f at p,

$$\kappa_\nu(p) := \frac{\langle \hat{\gamma}''(0), \nu(p) \rangle}{\|\hat{\gamma}'(0)\|^2} \neq 0.$$

ここは block の外へ移動

(cf. SUY 2009 for cuspidal edges)

Main result III: intrinsic criteria for singularity types

Theorem C

Let $f : \Sigma \to \mathbb{R}^3$ be a frontal, and let $p \in \Sigma$ be a 3-type edge. Assume that f is generic at p.
Then, the following conditions are equivalent:

(i) f is a $(3,4)$-cuspidal edge at p.

(ii) $\boxed{\beta(p) \neq 0}$ holds.

Moreover, the following conditions are equivalent:

(i) f is a $(3,5)$-cuspidal edge at p.

(ii) $\boxed{\beta \equiv 0 \quad \text{on } S(f) \text{ near } p, \qquad \delta(p) \neq 0}$ holds.

Main result III: intrinsic criteria for singularity types

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30 min

Application

Are these singularity types preserved by isometries?

To apply Theorem C, we need to know: Is being a 3-type edge intrinsic?

Towards preservation under isometries

Let $f : \Sigma \to \mathbb{R}^3$ be a frontal with ν, and let $p \in S(f)$ be a rank-1 singular point. Let $\lambda := \det(f_u, f_v, \nu)$ be the signed area density function.

Definition (singular point of degeneracy order n)

Let $n \geq 1$. A rank-1 singular point p has **degeneracy order** n if there exist smooth real-valued functions $\widehat{\lambda}$ and α such that

$$\lambda = \alpha\,\widehat{\lambda}^{\,n}, \qquad d\widehat{\lambda}(p) \neq 0, \qquad \alpha(p) \neq 0.$$

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- $n = 1$: the usual non-degenerate case (rank-1 is automatic).

Definition (first-kind singular point)

Let $S(f)$ be a regular curve near p. p is the first-kind if $T_pS(f) \neq \ker(df_p)$.

Intrinsic characterization of 3-type edges

Intrinsicness

The first-kind condition and degeneracy order n are both **intrinsic**.

Proposition

Let p be a rank-1 singular point of a frontal $f : \Sigma \to \mathbb{R}^3$. Then

$$\boxed{\begin{array}{ccc} p \text{ is of the first kind and} & & \\ p \text{ has degeneracy order 2} & \Longleftrightarrow & f \text{ has a 3-type edge at } p. \end{array}}$$

- Thus, 3-type edges can be viewed intrinsically.
- Since theorem C and this proposition, we obtain the following corollary.

Corollary: preservation of $(3,4)$- and $(3,5)$-cuspidal edges

For generic singular points of the first-kind with degeneracy order 2, isometric f, $\bar{f}$ have

$$\boxed{\text{same } ds^2} \implies \boxed{\text{same intrinsic data } \beta,\ |\delta|}$$
$$\implies \boxed{\text{being } (3,4)\text{- or } (3,5)\text{- is preserved by isometries}}.$$

Corollary D

Let

$$f : \Sigma \to \mathbb{R}^3, \qquad \bar{f} : \overline{\Sigma} \to \mathbb{R}^3$$

be locally isometric frontals near corresponding generic singular points p and $\overline{p}$ ~~of the first kind and degeneracy order 2~~.
Then, for each $j = 4, 5$, the following conditions are equivalent:

(i) f is a $(3,j)$-cuspidal edge at p.

(ii) $\bar{f}$ is a $(3,j)$-cuspidal edge at $\overline{p}$.

36min

Rigidity of $\mathcal{A}$-equivalence classes among generic frontals

For locally isometric generic frontals, is the singularity type preserved?

$$\boxed{f,\ \bar{f}:\text{isometric and generic frontals}} \Longrightarrow \boxed{f \sim_{\mathcal{A}} \bar{f}}\ ?$$

Singularity type	$f \sim_{\mathcal{A}} \bar{f}$ **?**	**Reference**
Cuspidal edge, swallowtail, cuspidal cross cap	Yes	Martins–Saji–Umehara–Yamada (2016) Honda–Naokawa–Umehara–Yamada (2020)
$(2,5)$-cuspidal edge	Yes	Honda–Saji (2019)
$(4,5)$- and $(4,5,+7)$-cuspidal edge	**No**	Hattori–Honda–Morimoto (2025)
$(3,4)$- and $(3,5)$-cuspidal edges	**Yes**	This work

Summary and outlook

New constructions

- NACS for ~~first-kind and degeneracy order 2 singular points.~~ 3-type edges
- Intrinsic ~~data~~ β and δ from ds^2. invariants
- Degeneracy order n for rank-1 singular points. frontal

Main results

(i) For a $(3,4)$-cuspidal edge,

$$\beta(p) > 0 \implies \text{convex}, \qquad \beta(p) < 0 \implies \text{saddle-shaped}.$$

(ii) K is bounded $\iff$ $\beta \equiv \delta \equiv 0$.

(iii) β, δ intrinsically characterize generic $(3,4)$- and $(3,5)$-cuspidal edges.

(iv) Isometries preserve these two singularity types in the generic class.

Outlook

- Intrinsic classification of non-generic 3-type edges.
- Geometric meaning of δ.

40 min

Appendix: main results at a glance

We introduce new intrinsic invariants β, δ.

First fundamental form ds^2 $\Longrightarrow$ Intrinsic data β, δ

Gaussian curvature	Local shape	Singularity type
K bounded	$\operatorname{sgn}\beta(p)$	β, δ are zero or not
$\Longleftrightarrow$ $\beta \equiv \delta \equiv 0$	$\Longrightarrow$ convex / saddle	$\Longrightarrow$ $(3,4)$ / $(3,5)$

Generic $(3,4)$ and $(3,5)$ are preserved by isometries.

Appendix: Singular and null directions

Let $f : \Sigma \to \mathbb{R}^3$ be a frontal. Assume that $S(f)$ is a regular curve consisting of rank 1 singular points near $p \in S(f)$.
Let $\gamma(t)$ parametrize $S(f)$ with $\gamma(0) = p$.

Definition Kokubu–Rossman–Saji–Umehara–Yamada (2005)

- The one-dimensional subspace

$$T_{\gamma(t)}S(f) = \mathrm{span}\{\gamma'(t)\}$$

  is called the **singular direction**.

- The one-dimensional subspace

$$\ker(df)_{\gamma(t)}$$

  is called the **null direction**. A nonzero vector $\eta(t)$ satisfying

$$df_{\gamma(t)}(\eta(t)) = 0$$

  is called a **null vector**.

Appendix: m-type edges are frontals

Fact

Every m-type edge is a frontal.

An m-type edge is $\mathcal{A}$-equivalent to

$$g(u,v) = \big(u, v^m, v^{m+1}\alpha(u,v)\big).$$

Set

$$\widetilde{\nu} = \Big(-mv^{m+1}\alpha_u,\ -v\{(m+1)\alpha + v\alpha_v\},\ m\Big).$$

Then,

$$\langle g_u, \widetilde{\nu}\rangle = \langle g_v, \widetilde{\nu}\rangle = 0, \qquad \widetilde{\nu} \neq 0.$$

Therefore,

$$\nu := \frac{\widetilde{\nu}}{\|\widetilde{\nu}\|}$$

is a smooth unit normal vector field along g.

$$\boxed{m\text{-type edge} \quad \Longrightarrow \quad \text{frontal (preserved by } \mathcal{A}\text{-equivalent)}}$$

Appendix: Intrinsic nature of the assumptions

Let

$$ds^2 = E\,du^2 + 2F\,du\,dv + G\,dv^2.$$

- **First kind:** the null direction and the singular set $S(f)$ are determined by ds^2. Hence

  $$\eta(p) \notin T_pS(f)$$

  is intrinsic.
- **Degeneracy order** 2**:** Since

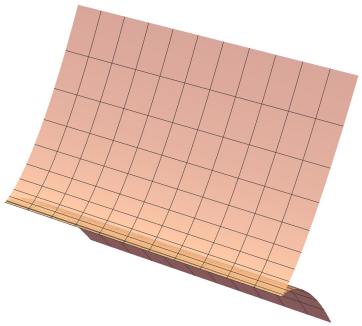
  $$\lambda^2 = EG - F^2,$$

  the transverse vanishing order of λ along the regular singular curve is determined by ds^2.

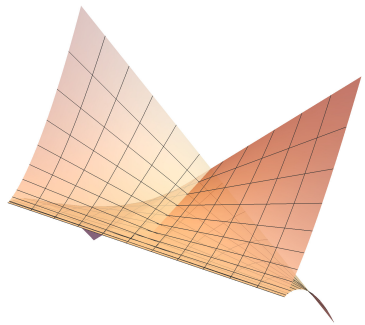
Both conditions are intrinsic.

Appendix: examples of first- and second-kind singularities

First-kind singularities

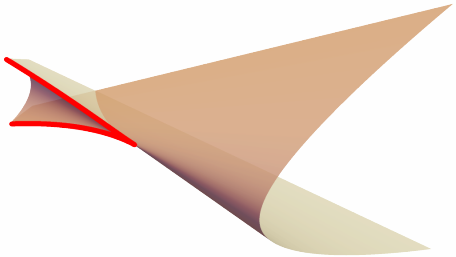


Cuspidal edge

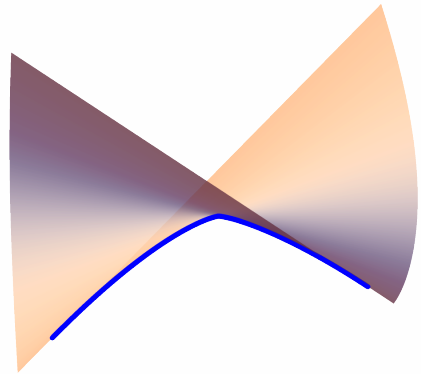


Cuspidal cross cap

Second-kind singularities at $p = (0, 0)$



Swallowtail



Cuspidal butterfly

Appendix

- Already for 4-type edges, the singularity type need not be preserved by isometries.
- Thus, it is nontrivial to ask whether this happens for 3-**type edges**.



$(4,5)$-cuspidal edge $(\sim_{\mathcal{A}} (u,\ v^4,\ v^5))$



$(4,5,+7)$-cuspidal edge $(\sim_{\mathcal{A}} (u,\ v^4,\ v^5 + v^7))$

Figure: Isometric surfaces with **different singularity types**.
(Hattori–Honda–Morimoto (2025))

Appendix: 2-type edge

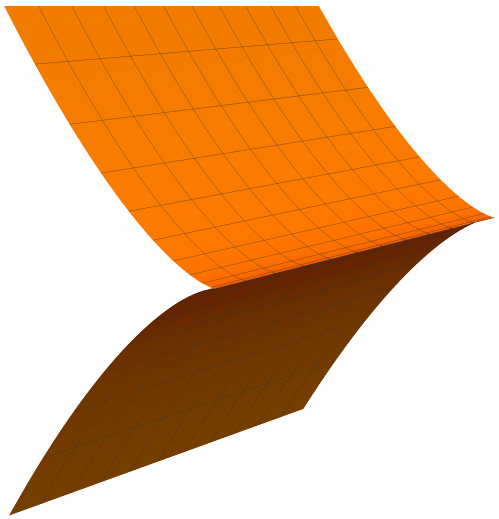
Relation to generalized cuspidal edges

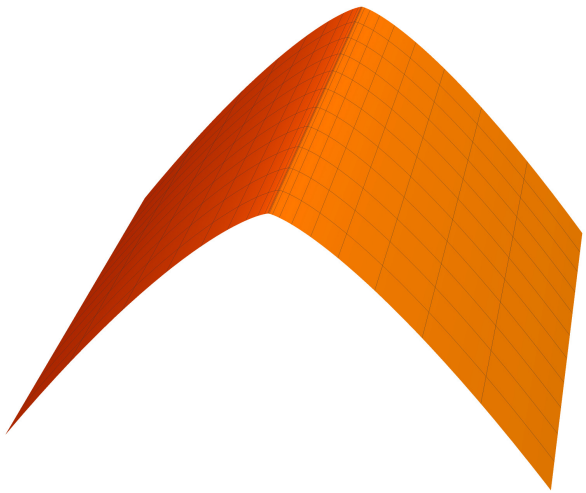
- For $m = 2$, m-type edge is exactly a **generalized cuspidal edge**: (Honda–Naokawa–Saji–Umehara–Yamada (2020))

$$(u, v) \mapsto (u, v^2, v^3\rho(u, v)).$$

- Thus, 3-type edge can be viewed as a **higher-order analogue**:

$$(u, v) \mapsto (u, v^3, v^4\rho(u, v)).$$





Appendix: Singular curvature

Let $\gamma(t)$ parametrize the singular set, and set

$$\widehat{\gamma} = f \circ \gamma, \qquad \widehat{\nu} = \nu \circ \gamma.$$

Singular curvature (Saji–Umehara–Yamada (2009))

For cuspidal edge,

$$\boxed{\kappa_s(t) := \mathrm{sgn}\big(d\lambda_{\gamma(t)}(\eta_{\gamma(t)})\big) \frac{\det\big(\widehat{\gamma}'(t), \widehat{\gamma}''(t), \widehat{\nu}(t)\big)}{\|\widehat{\gamma}'(t)\|^3}.}$$

The sign of κ_s is well-defined.

Appendix: Limiting normal curvature

Definition (limiting normal curvature and genericity)

Saji–Umehara–Yamada(2009); generalized by Martins–Saji–Umehara–Yamada(2016)

Let $f : \Sigma \to \mathbb{R}^3$ be a frontal, ν a unit normal vector field along f, and $p \in S(f)$ a singular point of the first kind.
Take an adapted coordinate system (u, v) such that

$$S(f) = \{v = 0\}, \qquad f_u(u, 0) \neq 0.$$

The **limiting normal curvature** is defined by

$$\kappa_\nu(u) := \frac{\langle f_{uu}(u, 0),\, \nu(u, 0)\rangle}{\|f_u(u, 0)\|^2}.$$

In particular, if

$$\kappa_\nu(p) \neq 0,$$

we say that f is **generic at** p.

Appendix: Signed limiting normal curvature

Let ξ be a singular vector field and η a null vector field satisfying

$$\eta f = \eta^2 f = \mathbf{0} \text{ on } S(f), \qquad \xi f,\ \eta^3 f \text{ linearly independent on } S(f).$$

New in this work

Definition

For a 3-type edge, define

$$\widetilde{\kappa}_\nu(p) := \frac{\det\left(\xi f, \eta^3 f, \xi^2 f\right)}{\|\xi f \times \eta^3 f\| \, \|\xi f\|^2}(p).$$

- $\widetilde{\kappa}_\nu$ is independent of the choice of a positively oriented pair (ξ, η).
- It is the ordinary limiting normal curvature with the normal orientation fixed by (ξ, η):

$$\widetilde{\kappa}_\nu = \pm \kappa_\nu.$$

Appendix: Cuspidal curvatures

Let ξ be a singular vector field and η a null vector field such that

$$\eta f = \eta^2 f = \mathbf{0} \text{ on } S(f), \qquad \xi f,\ \eta^3 f \text{ linearly independent on } S(f).$$

Definition ($(m, m+i)$-cuspidal curvatures)
Martins–Saji–Santos–Teramoto (2024)

The $(3,4)$-cuspidal curvature is defined by

$$\omega_{3,4}(p) := \frac{\|\xi f(p)\|^{4/3} \det\big(\xi f(p), \eta^3 f(p), \eta^4 f(p)\big)}{\|\xi f(p) \times \eta^3 f(p)\|^{7/3}}.$$

The $(3,5)$-cuspidal curvature is defined by

$$\omega_{3,5}(p) := \frac{\|\xi f(p)\|^{5/3} \det\big(\xi f(p), \eta^3 f(p), \eta^5 f(p)\big)}{\|\xi f(p) \times \eta^3 f(p)\|^{8/3}}.$$

- $\omega_{3,4}(p)$ is an invariant (NOT intrinsic).
- If $\omega_{3,4} \equiv 0$ along the singular set, then $\omega_{3,5}(p)$ is also an invariant.

Appendix: Product curvatures and β, δ

The cuspidal curvatures $\omega_{3,4}$ and $\omega_{3,5}$ were introduced by Martins–Saji–Santos–Teramoto.

New in this work

Definition (product curvatures in this work)

$$\Pi_{3,4} := \widetilde{\kappa}_\nu \omega_{3,4}, \qquad \Pi_{3,5} := \widetilde{\kappa}_\nu \omega_{3,5}.$$

$\Pi_{3,5}$ is well-defined when $\omega_{3,4} \equiv 0$ along the singular set.

Relation with β, δ

$$\boxed{\Pi_{3,4} = \frac{1}{2}\beta}, \qquad \boxed{\beta \equiv 0 \ \&\ \text{generic} \implies \Pi_{3,5} = \frac{1}{2}\delta}.$$

δ is defined without assuming $\omega_{3,4} \equiv 0$.

Appendix: from cuspidal curvatures to intrinsic data

For 3-type edges, Martins–Saji–Santos–Teramoto (2024) introduced the cuspidal curvatures

$$\omega_{3,4}, \qquad \omega_{3,5}.$$

These quantities measure higher-order cuspidal behavior.

- $\omega_{3,4}(p)$ is an invariant.
- If $\omega_{3,4} \equiv 0$ along the singular set, then $\omega_{3,5}(p)$ is also an invariant.

Key relations

Under the genericity assumption,

$$\boxed{\omega_{3,4}(p) \neq 0 \quad \Longleftrightarrow \quad \beta(p) \neq 0.}$$

If $\beta \equiv 0$ along the singular set near p, then

$$\boxed{\omega_{3,5}(p) \neq 0 \quad \Longleftrightarrow \quad \delta(p) \neq 0.}$$

Appendix: non-intrinsic criteria for $(3,4)$- and $(3,5)$-cuspidal edges

New in this work

Criterion for a $(3,4)$-cuspidal edge

Let $f : \Sigma \to \mathbb{R}^3$ be a frontal and $p \in \Sigma$. Then

$$p \text{ is a } (3,4)\text{-cuspidal edge} \iff \begin{cases} p \text{ is a 3-type edge,} \\ \omega_{3,4}(p) \neq 0 \end{cases}$$

Criterion for a $(3,5)$-cuspidal edge

Let $f : \Sigma \to \mathbb{R}^3$ be a frontal and $p \in \Sigma$. Then

$$p \text{ is a } (3,5)\text{-cuspidal edge} \iff \begin{cases} p \text{ is a 3-type edge} \\ \omega_{3,4} \equiv 0 \quad \text{on } S(f) \text{ near } p, \\ \omega_{3,5}(p) \neq 0. \end{cases}$$

These criteria are not intrinsic.

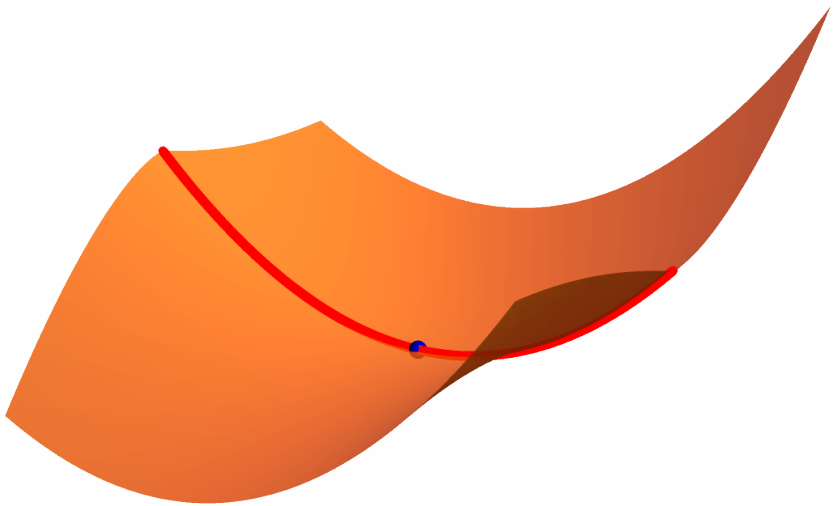
Appendix: examples of Main result III

$(3,4)$-**cuspidal edge**

$$f_{34}(u,v) = \left(u,\ v^3,\ \frac{u^2}{2} + uv^4 + v^5\right).$$

For $p = (u_0, 0)$ with $u_0 \neq 0$,

$$\boxed{\beta(p) \neq 0}.$$

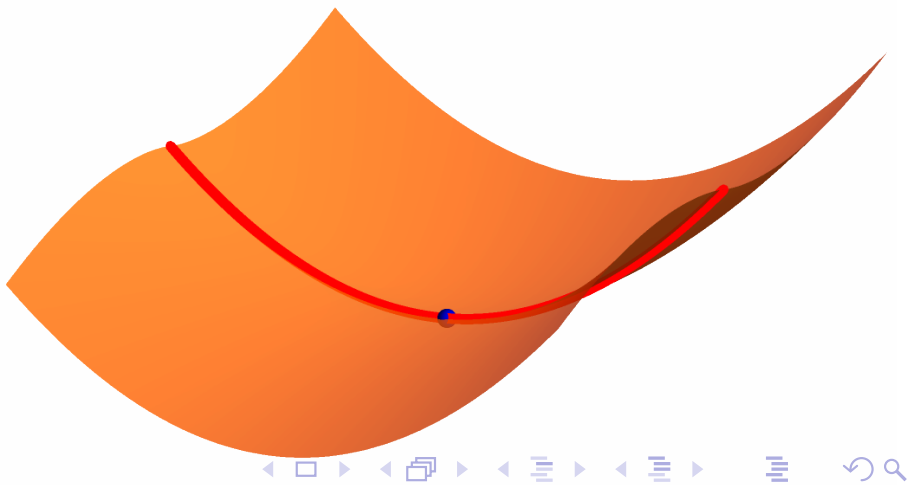


$(3,5)$-**cuspidal edge**

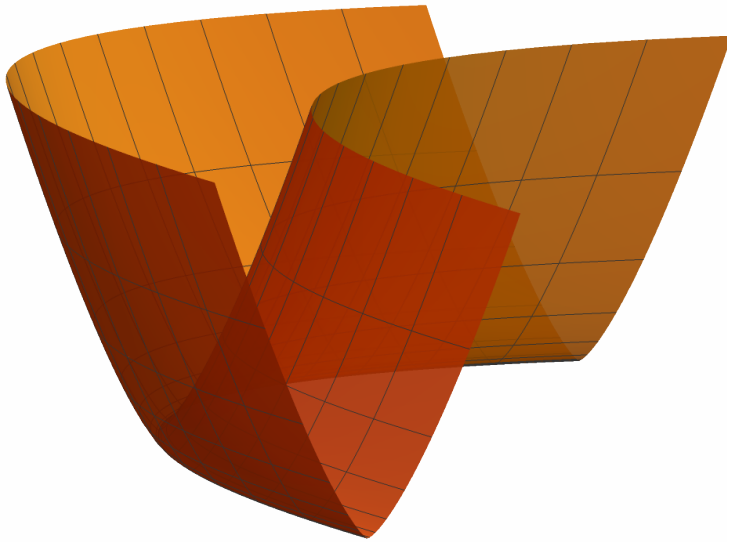
$$f_{35}(u,v) = \left(u,\ v^3,\ \frac{u^2}{2} + v^5\right).$$

For $p = (u_0, 0)$,

$$\boxed{\beta \equiv 0 \text{ on } S(f), \quad \delta(p) \neq 0}.$$



Appendix: why genericity is needed in Theorem C



$$f(u,v) = \left(u, \frac{u^2}{2} + \frac{v^3}{6}, \frac{v^4}{24}\right)$$

$(3,4)$-cuspidal edge, but $\kappa_\nu \equiv 0$ and $\beta \equiv 0$ along $S(f)$.

- In theorem C (and proposition B), the assumption $\kappa_\nu(p) \neq 0$ is essential. $(\kappa_\nu(p) = 0 \implies \beta(p) = 0)$
- **Without genericity, $\beta(p)$ does not characterize $(3,4)$-cuspidal edges**.

Appendix: Transformation formulas between NACSs

Let (u, v) and (ξ, η) be two NACSs near p, and write the coordinate transformation as $u = u(\xi, \eta), \quad v = v(\xi, \eta)$.
Let $\widehat{E}$ and $\widehat{G}$ denote the coefficients of the first fundamental form with respect to (ξ, η).

Higher-order transformation formulas between NACSs

Along the singular set $\{v = 0\}$ (equivalently, $\{\eta = 0\}$),

$$
\begin{aligned}
\widehat{E}_{\eta^3} &= v_\eta E_{v^3}, \\
\widehat{E}_{\eta^4} &= E_{v^4} + 6 v_{\eta^2} E_{v^3}, \\
\widehat{E}_{\eta^5} &= v_\eta E_{v^5} + 10 v_{\eta^2} v_\eta E_{v^4} + \left(10 v_{\eta^3} + 15 \{v_{\eta^2}\}^2 v_\eta\right) E_{v^3}, \\
\widehat{G}_{\eta^5} &= v_\eta \left(G_{v^5} + 120 v_{\eta^2}\right), \\
\widehat{G}_{\eta^6} &= G_{v^6} + 27 v_{\eta^2} G_{v^5} + 300 v_\eta v_{\eta^3} + 1170 \{v_{\eta^2}\}^2.
\end{aligned}
$$

Here $\widehat{E}$, $\widehat{G}$, and the derivatives of v are evaluated at $(\xi, 0)$, while E and G are evaluated at $(u(\xi, 0), 0)$.

Appendix: A sign change of β in the generic case

The surface is generic along the whole singular set, but the origin is not a $(3,4)$-cuspidal edge.

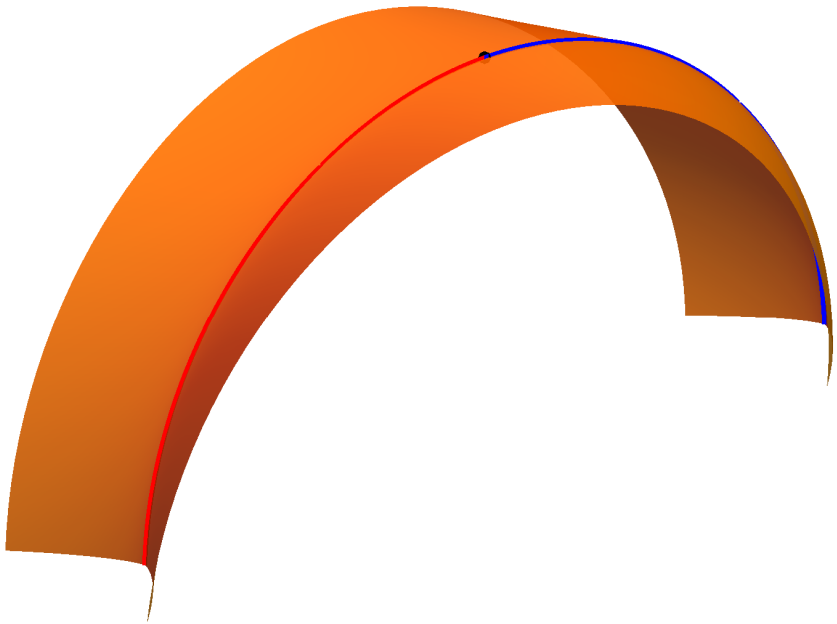


Figure: A surface with $\beta(u) = u$. It has a $(3,4)$-cuspidal edge for $u \neq 0$.

Appendix: A sign change of β at a non-generic point

The surface has a $(3,4)$-cuspidal edge at every point of the singular set, but the origin is non-generic.

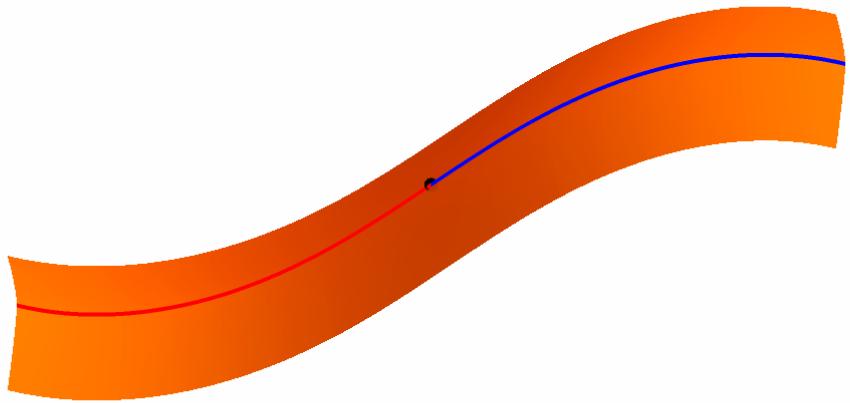


Figure: A $(3,4)$-cuspidal edge with $\beta(u) = u$. The origin is non-generic.

Appendix: A non-generic example with a non-planar singular curve

Let the singular set be $S(f) = \{v = 0\}$.
An example that is non-generic along the singular set and whose singular curve is **non-planar**.

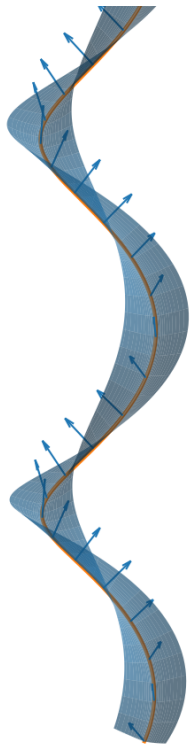


Figure: An example of a $(3, 4)$-cuspidal edge with $\beta = 0$.

Appendix: Map formulas for β and δ

Let (u, v) be a local coordinate system satisfying

$$S(f) = \{v = 0\}, \quad f_v(u, 0) = f_{vv}(u, 0) = \mathbf{0}, \quad \mathrm{rank}(f_u, f_{v^3})(u, 0) = 2.$$

For δ, we fix the orientation of ∂_v.

Map formulas for β and δ

The quantities β and δ, originally defined in a NACS, can be written as follows:

$$\beta(u) = 2\frac{\det(f_u, f_{v^3}, f_{uu})\det(f_u, f_{v^3}, f_{v^4})}{\|f_u\|^{2/3}\|f_u \times f_{v^3}\|^{10/3}}(u, 0)$$

$$\begin{aligned}\delta(u) = {} & 2\frac{\det(f_u, f_{v^3}, f_{uu})\det(f_u, f_{v^3}, f_{v^5})}{\|f_u\|^{1/3}\|f_u \times f_{v^3}\|^{11/3}}(u, 0) \\ & - \frac{10}{3}\frac{\|f_u\|^{5/3}}{\|f_u \times f_{v^3}\|^{17/3}}\left\{\langle f_{v^3}, f_{v^4}\rangle - \frac{\langle f_u, f_{v^3}\rangle\langle f_u, f_{v^4}\rangle}{\|f_u\|^2}\right\}\det(f_u, f_{v^3}, f_{uu})\det(f_u, f_{v^3}, f_{v^4})(u, 0) \\ & - \frac{4}{3}\frac{\|f_u\|^{5/3}}{\|f_u \times f_{v^3}\|^{17/3}}\left\{\langle f_{uu}, f_{v^3}\rangle - \frac{\langle f_u, f_{uu}\rangle\langle f_u, f_{v^3}\rangle}{\|f_u\|^2}\right\}\{\det(f_u, f_{v^3}, f_{v^4})\}^2(u, 0)\end{aligned}$$